Master Thesis

Uncertainty Quantification in Deep Reinforcement Learning: Systematic Benchmarks, Noise Paradox, and Calibration Complexity

by

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Uncertainty Quantification in Deep Reinforcement Learning: Systematic Benchmarks, Noise Paradox, and Calibration Complexity

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Abstract. Deep Reinforcement Learning (Deep RL) has demonstrated strong performance across domains, yet its adoption in safety-critical applications remains limited by the lack of reliable uncertainty quantification (UQ). This thesis presents a controlled benchmark of three representative methods—Quantile Regression DQN (QR-DQN), Bootstrapped DQN, and Monte Carlo Dropout DQN—evaluated in the LunarLander-v3 environment under five levels of observation noise, with 150 independent training runs and a unified suite of UQ metrics.
Results show that QR-DQN consistently achieves the highest UQ quality across CRPS, ACE, WIS, and coverage, confirming the superiority of distributional learning. A method-specific "noise paradox" is identified, where moderate noise improves Bootstrapped DQN by enhancing ensemble diversity. Calibration experiments further reveal method-dependent effects: ensemble- and dropout-based methods benefit consistently, while QR-DQN, already close to well-calibrated, exhibits mixed responses.
These findings contribute to a standardized evaluation framework, provide empirical evidence for distributional learning advantages, reveal method-specific noise responses, and show the conditional value of calibration. This work offers insights and guidance for applying Deep RL in uncertainty-sensitive applications.

Keywords: Uncertainty Quantification, Deep Reinforcement Learning, Distributional Learning, Ensemble Methods, Calibration, Safety-Critical Systems

1 Introduction

1.1 Background and Motivation

Deep Reinforcement Learning (DRL) has demonstrated remarkable progress in safety-critical domains such as autonomous driving [1, 2], medical diagnosis [3, 4], and financial trading [5, 6], demonstrating strong potential for deployment in practice. Despite these advances, uncertainty quantification (UQ) remains a major obstacle for real-world applications. Most deep RL approaches provide only point estimates without conveying the reliability of their predictions, which poses serious risks in safety-critical contexts [7, 8]. UQ methods are designed to provide

confidence measures for model outputs, thereby helping practitioners assess prediction reliability. However, applying UQ in DRL is particularly challenging due to error accumulation across sequential decisions, non-stationary environments, and the need to balance exploration with exploitation [9, 10].

While researchers have developed various UQ methods for deep RL in recent years [7–9, 11], current work still faces two persistent limitations:

Limited systematic comparisons: Most studies aim to improve individual UQ methods in isolation, rather than conducting comprehensive comparisons across approaches [7, 9, 11]. The lack of standardized, large-scale benchmarks aggravates reproducibility challenges in Deep RL [12, 13], making it difficult to fairly evaluate methods across diverse settings.

Insufficient attention to environmental factors: Many studies evaluate UQ methods under highly idealized conditions, overlooking how real-world perturbations influence performance. While in practice, agents must operate under noisy sensors, stochastic rewards, and dynamic environments, where poor uncertainty estimation can translate into catastrophic errors at critical moments [10, 14, 15]. Yet systematic investigations remain scarce.

To address these gaps, this paper investigates the following research questions:

1. How do widely used UQ methods in Deep RL perform under a unified experimental framework? What calibration behavior do they exhibit across multiple evaluation metrics?
2. How does observation noise affect the uncertainty estimation quality of these methods?

To answer these questions, this study implements a controlled experimental design in the LunarLander-v3 environment with 3 UQ methods $\times$ 5 noise levels (ranging from 0.000 to 0.100) $\times$ 10 random seeds, yielding 150 independent training runs. Performance is evaluated using four complementary metrics—CRPS (Continuous Ranked Probability Score), ACE (Average Calibration Error), WIS (Weighted Interval Score), and Coverage—which together provide a comprehensive assessment of UQ quality.

1.2 Main Contributions and Findings

Standardized evaluation framework. We introduce a reproducible benchmarking pipeline that fixes architectures, hyperparameters, training schedules, and evaluation windows. A full-factorial $3 \times 5 \times 10$ design (methods $\times$ noise levels $\times$ seeds) yields 150 independently trained agents with consistent episode-level logging and a unified metric suite, establishing reliable baselines for future studies.

Superiority of distributional learning. Under this unified protocol, QR-DQN consistently outperforms Bootstrapped DQN and MC Dropout DQN across

all uncertainty quality metrics, achieving substantially better scores and coverage closer to the nominal 90% target. This provides robust empirical evidence for the advantages of distributional learning among value-based UQ methods.

Noise paradox in ensembles. Bootstrapped DQN exhibits a counterintuitive non-monotonic noise response: moderate observation noise ($\sigma \approx 0.025$) improves UQ quality relative to both lower and higher noise levels. A finer-grained sweep ($\sigma \in [0.020, 0.030]$) confirms an optimum near 0.025, indicating that mild perturbations can enhance ensemble diversity without degrading task solvability.

Method-dependent calibration complexity. A two-parameter post-hoc calibration (bias shift Δ and variance scaling τ) yields consistent ACE and coverage gains for ensemble and dropout methods, but mixed effects for QR-DQN, which is already close to well-calibrated. In some settings, QR-DQN's CRPS worsens post-calibration despite ACE gains, underscoring the need for adaptive, method-aware calibration strategies.

2 Related Work

2.1 Background and Evolution

The study of uncertainty in reinforcement learning has progressed from simple exploration heuristics to dedicated modeling methods. It originated in the exploration–exploitation trade-off of multi-armed bandits, where Thompson sampling [16] and the Upper Confidence Bound (UCB) algorithm [17] established the theoretical foundation for using uncertainty to improve learning efficiency [18]. At this stage, uncertainty was treated mainly as a tool for exploration.

With the rise of deep reinforcement learning, the limitations of traditional ε-greedy strategies in high-dimensional state spaces became evident [19]. Research shifted from using uncertainty solely for exploration to evaluating the quality of the estimates themselves—transforming UQ from an auxiliary device into an independent research objective [20]. This perspective is particularly important in decision-support applications (e.g., clinical tasks requiring calibrated confidence) and offline RL, where reliable uncertainty estimates are essential in the absence of active exploration.

2.2 Mainstream Approaches

Uncertainty quantification (UQ) in reinforcement learning is commonly grouped into three families: distributional reinforcement learning, ensemble methods, and Bayesian approximations, which differ in their theoretical foundations, mechanisms, and limitations.

Distributional Reinforcement Learning Distributional RL models the full return distribution rather than point estimates. Bellemare et al. [21] introduced the C51 algorithm by extending the Bellman equation into distributional space:

$$\mathcal{Z}(s,a) \stackrel{d}{=} R(s,a) + \gamma \mathcal{Z}(S', A'). \tag{1}$$

Table 1: Mainstream UQ approaches in RL: assumptions, advantages, and limitations. (Type: A=aleatoric, E=epistemic).

Approach	**Type**	**Assumption**	**Advantages / Limitations**
Distributional RL	A	Learn return distribution	+ proper-scoring alignment – weak in epistemic uncertainty
Ensemble methods	E	Disagreement ≈ epistemic	+ exploration via disagreement – diversity/calibration issues
Bayesian methods	Both	Dropout ≈ variational posterior	+ low overhead – biased/unstable intervals

Dabney et al. [11] further proposed QR-DQN, which removed C51's fixed-support limitation through quantile regression:

$$\mathcal{L}_\tau(u) = u \cdot (\tau - \mathbb{I}\{u < 0\}), \tag{2}$$

where τ is the target quantile. This formulation flexibly captures arbitrary distributional shapes. While QR-DQN models returns accurately, it mainly reflects aleatoric rather than epistemic uncertainty [22].

Ensemble Methods Ensemble methods estimate uncertainty by aggregating predictions from multiple models. Osband et al. [9] proposed Bootstrapped DQN, which shares a feature extractor across multiple Q-heads trained on bootstrap samples. Disagreement among heads approximates epistemic uncertainty:

$$\sigma^2_{\text{epistemic}}(s,a) = \frac{1}{K}\sum_{k=1}^{K}\left(Q_k(s,a) - \bar{Q}(s,a)\right)^2. \tag{3}$$

This method improves exploration in Atari games with sparse rewards [9], but calibration stability remains uncertain across tasks.

Bayesian Methods Bayesian approaches maintain posterior distributions over parameters. Gal and Ghahramani [7] proposed Monte Carlo Dropout, interpreting dropout as a variational approximation: keeping dropout active at inference and averaging stochastic passes yields predictive distributions. However, Osband et al. [23] showed that MC Dropout biases confidence intervals in regression tasks, and temporal dependencies further challenge its assumptions [24]. Despite these issues, Bayesian methods can capture both aleatoric and epistemic uncertainty within a single framework.

Comparative Summary. Table 1 highlights that while each mainstream approach offers unique advantages, none provides a complete solution.

Transition to Comparative Studies. Existing evaluations remain fragmented: distributional methods' theoretical appeal does not always yield empirical gains, and ensemble and Bayesian methods often suffer from unstable calibration. More importantly, prior studies typically assess UQ indirectly via reward or exploration efficiency, rather than through dedicated uncertainty metrics. This gap motivates our systematic comparative analysis.

2.3 Limitations and Structural Gaps

Despite significant methodological advances, systematic empirical comparisons of UQ methods in RL remain scarce and often yield inconsistent conclusions. Osband et al. [9] focused on exploration efficiency, demonstrating the advantage of Bootstrapped DQN across 57 Atari games, but did not directly evaluate the calibration of uncertainty estimates. Later work [23] reported that MC Dropout produces systematically biased confidence intervals in regression tasks. Dabney et al. [11] showed that QR-DQN can accurately model return distributions, yet Mavrin et al. [25] argued that this accuracy does not necessarily translate into improved exploration or policy performance. These inconsistencies highlight that policy return or exploration efficiency alone is insufficient to evaluate the quality of uncertainty estimates. More direct and principled UQ metrics are needed to enable fair comparison.

A second limitation is the lack of systematic study on the impact of observational noise. Statistical learning theory predicts that noise increases task difficulty and degrades the quality of estimates [26], yet RL research has rarely validated this prediction empirically. Prior studies have primarily emphasized robust policy learning [27] or domain randomization for transfer [28], rather than directly examining how noise affects uncertainty quantification. Understanding this gap is critical for safety-critical applications where reliable confidence estimates are essential.

Finally, there is no widely adopted benchmark suite that integrates diverse environments, unified metric sets, consistent protocols, and reference baselines for UQ in RL [12, 13, 29]. As a result, methodological differences such as architecture choices, hyperparameter settings, and implementation details continue to undermine reproducibility and hinder cumulative progress. Reproducible evaluation pipelines are therefore essential to establish fair and reliable comparisons across UQ paradigms.

Recent advances (2023–2025) further highlight this gap. For instance, policy-gradient methods with built-in uncertainty estimation [30, 31] and probabilistic actor–critic variants [32] demonstrate that UQ research is expanding beyond value-based methods. Meanwhile, benchmarks such as UCRL-Bench [33] and uncertainty-focused Gym extensions [34] have been proposed to standardize evaluation protocols. However, these studies remain fragmented, reinforcing the need for the controlled framework introduced in this thesis.

In summary, prior work has advanced distributional, ensemble, and Bayesian approaches, but remains constrained by limited comparability, the absence of systematic noise studies, and the lack of standardized benchmarks. This thesis

addresses these gaps by establishing a controlled evaluation framework, systematically analyzing the role of observational noise, and providing a reproducible pipeline for fair and repeatable comparison of UQ methods in RL.

3 Methodology

This chapter presents the experimental methodology, filling the gaps discussed in Section 2 through a standardized evaluation framework. It enables fair and systematic comparison of the three UQ paradigms while controlling for artifacts and inconsistencies in implementation and evaluation.

3.1 Evaluation Framework

We design a standardized evaluation pipeline that enables reproducible and fair comparison of UQ methods. The framework is structured as a sequential workflow, from model training to uncertainty evaluation. Each stage addresses common methodological shortcomings in existing UQ literature, such as lack of reproducibility, inconsistent calibration, or insufficient metric design.

Pipeline Structure The evaluation follows a six-stage workflow:

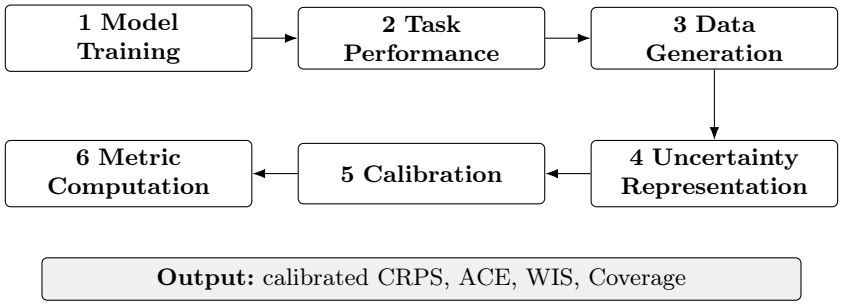

Fig. 1: Overview of the standardized evaluation pipeline. The workflow ensures fair, reproducible, and method-agnostic comparison across UQ paradigms.

Step 1. Model Training. All models are trained under identical network architectures and hyperparameters. We evaluate three UQ methods across five noise levels and ten random seeds, yielding 150 independently trained agents. This strict standardization ensures that differences are attributable solely to the UQ method.

Step 2. Data Generation. Each trained agent is evaluated over 100 episodes. Complete state–action–reward trajectories are recorded and organized at the episode level, forming the dataset for subsequent calibration and evaluation.

Step 3. Task Performance. Before uncertainty analysis, we verify that agents reliably solve the task by checking mean rewards and success rates. This guarantees that UQ quality is assessed only for agents that have learned stable policies.

Step 4. Uncertainty Representation. We extract method-specific uncertainty outputs: quantile distributions (QR-DQN), ensemble heads (Bootstrapped DQN), and MC samples (MC Dropout). All outputs are standardized into a unified 99-quantile representation for comparability.

Step 5. Calibration. A two-parameter calibration corrects systematic biases: bias shift (Δ) and variance scaling (τ). For each agent, episodes are split 80/20: the calibration set tunes parameters via grid search, and the held-out set prevents data leakage.

Step 6. Metric Computation. We compute four complementary UQ metrics: CRPS (distributional accuracy), ACE (calibration quality), Coverage (interval reliability), and WIS (sharpness–reliability trade-off).

Experimental Design The overall design follows a full factorial structure: 3 UQ methods $\times$ 5 noise levels $\times$ 10 random seeds, yielding 150 trained agents. Every agent undergoes the same training, evaluation, calibration, and metric computation protocol. Organizing data at the episode level ensures statistical independence in calibration and testing, while comprehensive logging of seeds, hyperparameters, and environment settings guarantees full reproducibility.

3.2 Experimental Environment and Noise Modeling

Task Environment Selection We use **LunarLander-v3** as the evaluation environment. It provides a well-established benchmark in reinforcement learning research [35] and has been used in prior studies of risk-sensitive and uncertainty-aware RL [36, 37]. The environment offers a continuous 8-dimensional state space and four discrete actions, and defines a clear success criterion (reward > 200). These properties make it a practical testbed for systematic evaluation of uncertainty quantification methods, while keeping computational requirements tractable for large-scale experiments.

Noise Modeling Design To systematically examine the robustness of UQ methods under different levels of observational uncertainty, we inject isotropic Gaussian noise into the state observations:

$$\tilde{s}_t = s_t + \mathcal{N}(0, \sigma^2 I), \quad s_t \in \mathbb{R}^8. \tag{4}$$

The noise levels were set as $\sigma \in \{0.000, 0.025, 0.050, 0.075, 0.100\}$. Pilot experiments showed that noise levels above $\sigma = 0.1$ substantially degraded task performance across all methods, and hence we excluded higher noise levels. Considering the typical range of state variables, $\sigma = 0.05$ corresponds to roughly $\sim$5% relative noise, which is comparable to common sensor error magnitudes [38].

3.3 Uncertainty Quantification Methods

Having established our experimental environment, we now present the three UQ methods that address uncertainty estimation through distinct paradigms:

QR-DQN (Distributional Method) QR-DQN [11] represents uncertainty through quantile distributions, learning the full return distribution rather than point estimates. The method partitions the return distribution into quantiles and minimizes the quantile regression loss, naturally providing both mean predictions and uncertainty bounds. This approach offers theoretical guarantees through proper distributional learning and efficient computation via quantile regression.

Bootstrapped DQN (Ensemble Method) Bootstrapped DQN [9] estimates uncertainty through ensemble disagreement, using multiple Q-heads trained on different bootstrap samples of the experience replay buffer. Each head shares a common feature extractor but maintains independent Q-value outputs. During training, bootstrap masks determine which heads update on a given experience, creating diversity across ensemble members [23]. Uncertainty arises from the variance across ensemble members, enabling the capture of both aleatoric and epistemic uncertainty through model diversity.

MC Dropout DQN (Bayesian Approximation) MC Dropout [7] approximates Bayesian inference by treating dropout as a variational inference technique, interpreting dropout training as approximate variational inference in deep Gaussian processes. By keeping dropout active during inference and averaging predictions over multiple stochastic forward passes, the method yields an approximate posterior distribution over network parameters. This approach provides a computationally efficient alternative to full Bayesian methods while producing principled uncertainty estimates.

3.4 Implementation Details

Framework and Design All experiments were conducted using the **RL Baselines3 Zoo** [35] framework, which provides community-validated hyperparameter configurations for fair comparison. To balance reliability with experimental control, we adopted a hybrid implementation strategy: QR-DQN was implemented using the mature `QRDQN` module from `sb3-contrib`, while Bootstrapped DQN and MC Dropout DQN were custom-implemented within SB3 to ensure architectural consistency and unified evaluation protocols.

The custom implementations closely follow the original algorithmic specifications. Bootstrapped DQN replicates the multi-head architecture and bootstrap mask sampling described in [9], while MC Dropout follows the variational inference framework of [7] with dropout layers active during inference. Both implementations remain compatible with the SB3 training pipeline and

enable method-specific uncertainty extraction. All methods share identical network backbones (MLP with [512, 512] hidden layers) and training hyperparameters, ensuring that observed differences are attributable solely to the uncertainty quantification approach.

Method-Specific Parameters

QR-DQN. We use 170 uniformly spaced quantiles covering $(0.01, 0.99)$, following the fine-tuned configuration from **RL Baselines3 Zoo** [35]. This setup, validated in prior large-scale benchmarks, balances distributional granularity with computational efficiency. For standardized comparison across methods, the quantiles are linearly interpolated to 99 during evaluation.

Bootstrapped DQN. Following [9], our implementation uses 10 independent Q-heads with a shared feature extractor. During training, each experience sample is assigned to heads via bootstrap masks drawn from Bernoulli distributions with probability $p = 0.65$, consistent with both our configuration files and prior recommendations [23]. To prevent all-zero masks, such samples are assigned to a randomly selected head [9]. The choice of 10 heads balances uncertainty estimation quality with computational overhead, aligning with common practice in ensemble methods.

MC Dropout DQN. Dropout layers with rate $p = 0.1$ are inserted in the hidden layers and remain active during inference, following empirical guidelines for variational dropout in deep reinforcement learning [7]. At inference, 30 Monte Carlo forward passes are performed to approximate the posterior distribution. This sampling budget balances estimation accuracy with computational cost and is consistent with established practices in uncertainty estimation for deep RL [39].

Reproducibility Measures All experiments use fixed random seeds and deterministic algorithms where possible. Each of the 150 trained models was processed independently with full state preservation for subsequent evaluation phases, ensuring complete reproducibility of results.

3.5 Evaluation Metrics

Evaluating the quality of uncertainty quantification (UQ) in RL requires multiple perspectives. No single metric can capture both distributional fidelity and calibration quality, and relying only on task reward, predictive variance, or negative log-likelihood (NLL) [9, 39] provides an incomplete picture. To this end, we adopt four complementary metrics grounded in *proper scoring rules* [40, 41]: CRPS for distributional accuracy, ACE and Coverage for calibration, and WIS for the sharpness–reliability trade-off. Task reward is additionally reported as a sanity check to confirm that UQ quality is not confounded by training failures.

CRPS. The Continuous Ranked Probability Score (CRPS) measures the global agreement between predictive distributions and outcomes:

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \big[F(z) - \mathbb{I}\{y \le z\}\big]^2 \, dz.$$

As a strictly *proper* score, CRPS rewards well-calibrated distributions and penalizes both bias and dispersion errors. It is particularly suitable for distributional methods such as QR-DQN [11]. Lower CRPS values indicate better fit, with zero denoting perfect prediction.

ACE. The Average Calibration Error (ACE) quantifies how well nominal confidence levels match empirical coverage across M bins:

$$\mathrm{ACE} = \tfrac{1}{M} \sum_{m=1}^{M} \big|\alpha_m - \widehat{\mathrm{cov}}_m\big|, \quad M = 10.$$

An ACE of zero implies perfect calibration. Positive deviations generally indicate overconfidence (intervals too narrow), whereas negative deviations reflect underconfidence. Since deep Q-learning tends toward overestimation bias [42], calibration analysis is especially relevant here.

Coverage. Coverage directly reports the empirical reliability of prediction intervals. For the 90% interval,

$$\mathrm{Coverage}_{90\%} = \tfrac{1}{N} \sum_{i=1}^{N} \mathbb{I}\{y_i \in [q_{0.05}, q_{0.95}]\}.$$

Ideally, the empirical frequency should approach the nominal confidence (0.90). Large deviations indicate miscalibration. While intuitive, Coverage alone does not penalize overly wide intervals, which can artificially inflate apparent reliability.

WIS. The Weighted Interval Score (WIS) overcomes this by balancing sharpness with coverage:

$$\mathrm{WIS}_\alpha = \tfrac{1}{N} \sum_{i=1}^{N} \Big[(q_{1-\alpha/2} - q_{\alpha/2}) + \tfrac{2}{\alpha} \cdot \mathrm{pen}_i\Big],$$

where pen_i penalizes observations outside the interval. Low WIS values correspond to narrow and accurate intervals, whereas overly wide or frequently invalid intervals yield higher scores. Unlike Coverage, WIS discourages trivial solutions that maximize coverage by inflating interval width.

All four metrics are computed after calibration (Section 3.6), ensuring that evaluation reflects intrinsic method quality rather than raw biases of Q-learning. Together, they provide a multidimensional foundation: CRPS captures overall distributional accuracy, ACE and Coverage assess calibration, and WIS enforces interval efficiency.

3.6 Calibration Protocol

Motivation and Procedure Before metrics evaluation, we introduce a calibration step to correct systematic biases in deep Q-learning. Calibration is not an evaluation metric but a preprocessing step: without it, CRPS, ACE, Coverage, and WIS would partly reflect artifacts of training rather than intrinsic uncertainty modeling. Deep Q-learning is known to suffer from overestimation bias [42], which distorts predictive uncertainty. To address this, we adopt a simple post-hoc calibration procedure, following practices in probabilistic forecasting and risk-sensitive learning [43–45].

Specifically, we employ a two-parameter adjustment that maps raw predictions (μ, σ) to calibrated estimates $(\mu_{\text{cal}}, \sigma_{\text{cal}})$: $\mu_{\text{cal}} = \mu + \Delta, \quad \sigma_{\text{cal}} = \sigma \times \tau$, where Δ corrects mean bias and τ rescales variance to avoid intervals that are too narrow or too wide. Such parameterized calibration is well established in probabilistic forecasting and Bayesian post-processing [44, 46]. Our procedure minimizes the Average Calibration Error (ACE) across multiple coverage levels. For distributional methods such as QR-DQN, calibration is applied element-wise to quantiles, preserving the structure of the predicted distribution.

Implementation Protocol Calibration is performed at the episode level to ensure independence and avoid temporal leakage. The procedure consists of four stages:

Data Collection. Each trained model is evaluated over 100 episodes, with complete state–action trajectories recorded. Empirical returns are computed as ground-truth targets:

$$Q^{\text{target}}(s_t, a_t) = \sum_{k=t}^{T} \gamma^{k-t} r_k. \tag{5}$$

Episode-level treatment reduces temporal correlation in sequential data [47, 48].

Episode Split. Episodes are randomly divided into an 80% calibration set and 20% validation set. Random splitting, rather than sequential partitioning, ensures statistical independence and prevents temporal leakage [49].

Parameter Optimization. The bias term is estimated as $\Delta = \mathbb{E}[Q^{\text{target}} - \mu]$. The scaling factor τ is optimized by grid search over $[0.1, 10.0]$ with 100 candidates, minimizing ACE across multiple coverage levels (50%, 80%, 90%). This is followed by bounded Nelder–Mead refinement [50, 51].

Validation and Evaluation. Calibrated parameters are applied to the held-out validation set for final metric computation. For QR-DQN, calibration is applied directly to each quantile, correcting systematic biases while maintaining the relative shape of the distribution.

3.7 Statistical Analysis Methods

Hypothesis Testing Scheme Since most metrics deviate from normality, we primarily adopt *non-parametric* methods. Pairwise method comparisons are conducted using the Mann–Whitney U test, while noise effects (paired) are analyzed with the Wilcoxon signed-rank test. Both *p-values* and *effect sizes* are reported to avoid over-reliance on significance alone.

Effect Sizes and Interval Estimation For approximately normal distributions, we report Cohen's d:

$$d = \frac{\mu_1 - \mu_2}{\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}}. \tag{6}$$

For distribution-free cases, we report Cliff's δ:

$$\delta = \frac{\#(x_i > y_j) - \#(x_i < y_j)}{n_1 n_2}. \tag{7}$$

Confidence intervals for key statistics are computed using the bootstrap percentile method with B=1000 resamples.

4 Results

We conducted 150 independent training runs in LunarLander-v3 (3 UQ methods $\times$ 5 noise levels $\times$ 10 seeds). A preliminary check verified that all agents achieved meaningful task performance (Appendix B.1), ensuring that uncertainty evaluation is based on stable policies.

4.1 Aggregate Uncertainty Quality

We first aggregate results across the five noise levels for each method ($n = 50$ runs). CRPS, WIS, ACE, and 90% coverage are reported in Table 2. Additional statistics and effect sizes are given in Appendix B.2.

Table 2: Aggregate UQ quality across five noise levels ($n = 50$ runs per method). Lower is better for CRPS/WIS/ACE; coverage is in percentage.

Method	CRPS	WIS	ACE	Coverage (90%)
QR-DQN	17.7 ± 8.7	150.2 ± 89.3	0.14 ± 0.12	83 ± 16
Bootstrapped DQN	23.6 ± 12.6	365.0 ± 227.8	0.50 ± 0.17	31 ± 22
MC Dropout DQN	26.5 ± 10.9	392.4 ± 171.2	0.49 ± 0.12	33 ± 14

Statistical tests. Mann–Whitney U tests confirmed that QR-DQN significantly outperforms both Bootstrapped DQN and MC Dropout DQN on all four metrics (all $p < 0.001$, medium–large effect sizes). For example, QR-DQN achieves 31% lower CRPS than Bootstrapped DQN ($d = 0.92$) and 45% lower than MC Dropout DQN ($d = 1.23$). Differences between Bootstrapped and MC Dropout DQN were mostly non-significant (Appendix B.2).

Exploratory variant. A QR-Bootstrapped DQN hybrid (distributional + ensemble; Appendix B.3) was unstable and failed to outperform either baseline, supporting our focus on the three main paradigms.

4.2 Noise-Level Analysis

We next analyze robustness under increasing observation noise. Figure 2 plots calibrated CRPS across noise levels with 95% bootstrap confidence intervals. To complement the figure, Tables 3–5 report per-noise values for CRPS, WIS, and ACE, providing detailed numerical comparisons. Coverage values are reported in Appendix B.4.

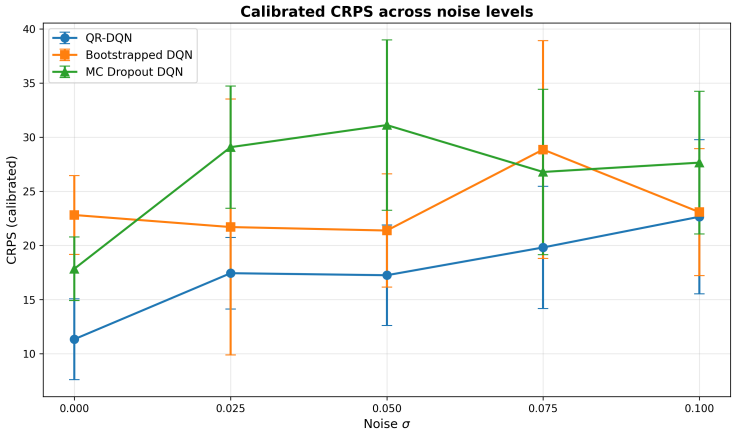

Fig. 2: Calibrated CRPS across observation noise levels ($\sigma = 0.0$–0.1). Each point shows the mean over 10 random seeds, with error bars denoting 95% bootstrap confidence intervals. Lower values indicate better uncertainty quality.

Key finding: the noise paradox. CRPS, WIS, and ACE all degrade monotonically for QR-DQN and MC Dropout DQN, confirming their sensitivity to noise. In contrast, Bootstrapped DQN exhibits a non-monotonic trend with improved scores at $\sigma = 0.025$—an effect we term the *noise paradox*. Fine-grained experiments ($\sigma = 0.020$–0.030) confirmed the optimum at $\sigma = 0.025$ (Appendix B.5). Coverage results are reported in Appendix B.4.

Table 3: Calibrated CRPS across noise levels (mean ± SD, $n = 10$ seeds per cell).

Method	0.0	0.025	0.05	0.075	0.1
QR-DQN	11.3 ± 6.0	17.4 ± 5.3	17.2 ± 7.5	19.8 ± 9.1	22.6 ± 11.5
Bootstrapped DQN	22.8 ± 5.9	21.7 ± 19.1	21.4 ± 8.4	28.9 ± 16.2	23.1 ± 9.5
MC Dropout DQN	17.8 ± 4.7	29.1 ± 9.1	31.1 ± 12.7	26.8 ± 12.3	27.7 ± 10.6

Table 4: Calibrated WIS across noise levels (mean ± SD, $n = 10$ seeds per cell).

Method	0.0	0.025	0.05	0.075	0.1
QR-DQN	101.8 ± 65.2	157.0 ± 66.5	142.6 ± 72.0	182.2 ± 89.3	257.3 ± 112.4
Bootstrapped DQN	380.4 ± 103.5	353.5 ± 353.6	315.6 ± 162.2	453.0 ± 280.5	322.9 ± 167.5
MC Dropout DQN	240.6 ± 90.7	454.0 ± 144.6	451.6 ± 178.5	440.0 ± 192.2	375.4 ± 122.6

4.3 Calibration Effects

Finally, we examine the effect of post-hoc calibration. Table 6 reports average percentage gains across noise levels. Bootstrapped and MC Dropout DQN improve consistently, whereas QR-DQN—already well calibrated—shows mixed responses.

Statistical tests. Paired t-tests confirmed significant improvements in CRPS ($t = 8.47$, $p < 10^{-14}$, $d = 0.59$), WIS ($t = 13.89$, $p < 10^{-30}$, $d = 0.96$), and ACE ($t = 19.32$, $p < 10^{-47}$, $d = 1.34$). These medium–large effect sizes show that calibration systematically enhances UQ quality.

Per-noise effects. Bootstrapped and MC Dropout DQN improved at every noise level. QR-DQN showed negative CRPS gains at $\sigma = 0.0$ and $\sigma = 0.1$ with unstable ACE response. Detailed values are provided in Appendix B.6.

Noise interaction. ANOVA indicated no significant interaction between noise and calibration benefit ($F = 1.21$, $p = 0.29$). Correlation analysis, however, revealed parameter adaptation: the bias-shift Δ decreased with noise ($r = -0.41$, $p < 0.01$) while the variance-scaling τ increased ($r = 0.38$, $p < 0.01$), suggesting that calibration compensates by widening predictive intervals under noisier conditions.

Table 5: Calibrated ACE across noise levels (mean ± SD, $n = 10$ seeds per cell). Lower is better.

Method	0.0	0.025	0.05	0.075	0.1
QR-DQN	0.08 ± 0.07	0.13 ± 0.10	0.13 ± 0.10	0.18 ± 0.13	0.19 ± 0.15
Bootstrapped DQN	0.68 ± 0.04	0.48 ± 0.16	0.47 ± 0.14	0.44 ± 0.22	0.42 ± 0.15
MC Dropout DQN	0.41 ± 0.09	0.54 ± 0.13	0.57 ± 0.12	0.52 ± 0.12	0.49 ± 0.11

Table 6: Calibration gains across five noise levels ($n = 50$ per method). Values are mean ± SD. CRPS/ACE: percentage reduction; coverage: pp increase.

Method	CRPS reduction (%)	ACE reduction (%)	Coverage change (pp)
Bootstrapped DQN	11.2 ± 37.9	29.7 ± 23.8	30 ± 20
MC Dropout DQN	12.9 ± 28.6	31.2 ± 16.7	30 ± 10
QR-DQN	-2.9 ± 41.3	21.6 ± 93.8	20 ± 20

5 Discussion

Our evaluation of 150 independent models demonstrates three central findings about uncertainty quantification in deep reinforcement learning: the superiority of distributional learning (QR-DQN), the unexpected noise paradox in Bootstrapped DQN, and the method-dependent nature of calibration gains. These results challenge common assumptions and provide new perspectives for both theoretical understanding and practical applications.

5.1 Underlying Mechanisms of Distributional Learning Advantages

The consistently strong performance of QR-DQN across all metrics may be explained by how well distributional learning aligns with the uncertainty quantification task. Ensemble methods estimate uncertainty indirectly through disagreement between multiple point estimates:

$$UQ_{\text{ensemble}} = f(\text{Var}[Q_{\theta_1}, Q_{\theta_2}, \ldots, Q_{\theta_K}]). \tag{8}$$

This approach inevitably discards information about the full distributional shape. By contrast, QR-DQN directly learns the quantile distribution $F_Q(r|s, a)$, retaining richer statistical information about the return distribution.

A further explanation lies in the relationship between QR-DQN's training objective and the CRPS evaluation metric. As shown by [40], the CRPS can be expressed in terms of quantile scores as

$$CRPS(F, y) = \int_0^1 QS_\tau(F^{-1}(\tau), y)\, d\tau, \tag{9}$$

where QS_τ is the quantile score, equivalent to the pinball loss. Training with a finite set of quantile levels $\{\tau_k\}_{k=1}^K$ therefore corresponds to a discrete approximation of this integral. QR-DQN leverages this property by jointly optimizing multiple quantile losses. The quantile Huber loss used in practice is a smooth approximation of the pinball loss, introduced for numerical stability [11]. This formal link clarifies why QR-DQN's training is naturally aligned with the evaluation metrics considered in this study.

It is important to note that the training and evaluation objectives are not identical: QR-DQN minimizes quantile-based discrepancies with a Bellman bootstrap target, whereas CRPS evaluates predictions against realized returns. Still,

both belong to the same family of strictly proper scoring rules and share the quantile-loss foundation. This alignment plausibly contributes to QR-DQN's comparatively well-calibrated performance observed in our experiments, while ensemble and dropout variants remained severely overconfident. Nevertheless, ensemble-based methods retain practical appeal: they are comparatively simple to implement and scale naturally with additional heads, which explains their continued use despite weaker theoretical alignment with CRPS.

5.2 The Noise Paradox in Bootstrapped DQN

The improvement of Bootstrapped DQN at $\sigma = 0.025$—where performance temporarily improves before degrading—appears to deviate from conventional expectations. This counterintuitive phenomenon suggests inherent limitations of bootstrap-based methods in deterministic environments.

In noise-free settings, deterministic state transitions cause bootstrap heads to converge to highly similar Q-functions. This excessive similarity undermines the ensemble's effectiveness, as the lack of diversity prevents robust uncertainty estimation. Introducing mild observation noise changes this dynamic: when $\sigma = 0.025$, stochastic perturbations in state observations expose different heads to slightly different effective training sequences. These differences are amplified during learning, leading to greater diversity and improved ensemble uncertainty representation. The critical factor is the balance of noise intensity: very low noise fails to induce meaningful diversity, while excessive noise disrupts learning. A small but nonzero noise level appears to provide the right trade-off.

This aligns with the well-known accuracy–diversity trade-off in ensemble learning [52]. Importantly, the beneficial role of noise is method-specific: QR-DQN degrades gradually in line with distributional robustness, while MC Dropout DQN proves highly fragile, with steep performance drops at even mild noise.

5.3 Analysis of Calibration Complexity

The calibration experiments highlight a crucial but often overlooked phenomenon: baseline calibration quality largely determines both the magnitude and the character of posterior improvements. For QR-DQN, which is already close to well-calibrated, ACE consistently improves, but CRPS shows mixed responses. This divergence indicates that additional calibration may distort the learned distribution, even as nominal coverage improves. Such drawbacks of post-hoc calibration have been observed in other domains [43, 44], but are rarely discussed in reinforcement learning.

In contrast, ensemble- and dropout-based methods start from a state of severe miscalibration. For these methods, post-hoc calibration provides consistent benefits across all metrics. This contrast demonstrates that calibration is not universally beneficial but method-dependent: well-calibrated methods may suffer from over-correction, while miscalibrated methods predictably benefit from strong adjustment.

These findings suggest that calibration strategies should be tailored to method-specific characteristics, rather than applied uniformly. More broadly, they challenge the assumption that all UQ methods benefit equally from calibration and instead point to the need for adaptive strategies that adjust correction strength according to baseline miscalibration levels.

5.4 Assessment of Limitations

Several limitations must be acknowledged when interpreting the results.

First, the study was conducted on a single environment (LunarLander-v3), which constrains the generalizability of results. While this environment represents a standard continuous control task, domains with different complexity or structural characteristics may yield different outcomes.

Second, our noise modeling was limited to Gaussian perturbations of observations. In practice, uncertainty arises from heterogeneous sources, including model uncertainty, sensor faults, and dynamic environmental changes. These factors may produce different effects than those observed here.

Third, the methodological scope was restricted to three representative approaches: distributional learning (QR-DQN), bootstrap ensembles (Bootstrapped DQN), and dropout-based Bayesian approximation (MC Dropout DQN). Other families, such as policy-gradient approaches, deep ensembles, and normalizing-flow techniques, were not considered. Their exclusion narrows the scope of our conclusions and highlights the need for future comparisons across methodological families.

Finally, the evaluation emphasized UQ quality metrics (CRPS, ACE, WIS) rather than computational efficiency. In real applications, the trade-off between accuracy and efficiency is context-dependent: in medical or financial decision-support, higher costs may be acceptable for reliability, whereas in online control, efficiency may dominate. This study therefore positions itself as a quality-focused benchmark, while integration of efficiency–quality trade-offs remains a key direction for aligning UQ evaluation with deployment needs.

6 Conclusion

This thesis presents a systematic evaluation of uncertainty quantification (UQ) methods in deep reinforcement learning, based on large-scale controlled experiments across multiple algorithms and noise conditions. The study provides empirical evidence for several key findings that advance both theoretical understanding and practical deployment of UQ in Deep RL.

6.1 Key Findings and Scientific Contributions

Standardized evaluation framework. We established a controlled, reproducible evaluation pipeline (Sec. 3), combining a unified uncertainty representation, post-hoc calibration, and proper scoring rule metrics. This framework enables fair, cross-method comparison of UQ quality.

Superiority of distributional learning. QR-DQN consistently achieved the strongest results across all metrics, maintaining reliable coverage and strong calibration where ensemble- and dropout-based methods fell short. This advantage reflects the natural alignment between distributional learning objectives and uncertainty quantification requirements.

Method-specific noise effects. We identified a counterintuitive "noise paradox," where moderate observational noise improved Bootstrapped DQN by enhancing ensemble diversity. In contrast, QR-DQN degraded smoothly with noise, while MC Dropout proved highly fragile. These results demonstrate that noise cannot be treated uniformly across UQ methods, but interacts in method-specific ways.

Calibration complexity. Calibration gains were strongly method-dependent. Ensemble and dropout methods, initially miscalibrated, benefited consistently from post-hoc adjustment, whereas QR-DQN—already near well-calibrated—showed mixed responses, with improvements in coverage but occasional distortions in distributional accuracy. This challenges the assumption of universal calibration strategies and highlights the need for adaptive, method-aware protocols.

6.2 Practical Deployment Guidance

The findings translate into actionable guidance for real-world applications: (1) Prefer distributional methods such as QR-DQN when reliable confidence intervals are essential, especially in safety-sensitive domains. (2) Apply strong calibration to ensemble and dropout methods, while using lighter adjustment—or none at all—for already well-calibrated methods like QR-DQN. (3) Treat environmental noise with nuance: moderate noise can improve ensemble diversity, though robustness varies across methods. Deployment decisions should balance uncertainty quality with computational efficiency, which was outside the scope of this study but remains critical in practice.

6.3 Future Research Directions

Future work should extend this framework to diverse RL environments and noise models to test the generality of our findings. Methodological advances may include noise-aware ensemble designs that exploit perturbations constructively, and adaptive calibration strategies that tailor correction strength to baseline miscalibration. Expanding evaluation to policy-based and emerging UQ techniques, while integrating computational efficiency into benchmarks, will provide a more comprehensive landscape of uncertainty quantification in deep reinforcement learning.

This thesis contributes a standardized evaluation framework that demonstrates distributional learning advantages, uncovers method-specific noise effects, and clarifies the conditional value of calibration for reliable UQ in safety-critical reinforcement learning systems.

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A Notation and Terminology

This appendix provides definitions for mathematical notation, abbreviations, and terminology used throughout this thesis.

A.1 Mathematical Notation

Reinforcement Learning

s_t, s State at time t (or general state)

a_t, a Action at time t (or general action)

r_t, r Reward at time t (or general reward)

$Q(s, a)$ Q-value function for state-action pair

Q^{target} Target Q-value from empirical returns

$\pi(a|s)$ Policy (action probability given state)

γ Discount factor

T Episode termination time

θ Neural network parameters

Uncertainty Quantification

F Cumulative distribution function (CDF)

$F_Q(r|s, a)$ Q-value return distribution

μ, σ Mean and standard deviation

$(\mu_{\text{cal}}, \sigma_{\text{cal}})$ Calibrated mean and std

Δ Bias correction parameter (calibration)

τ Quantile level or variance scaling parameter

q_τ Quantile at level τ

$q_{0.05}, q_{0.95}$ 5th and 95th percentiles

α Confidence level

α_m Nominal confidence in bin m

$\widehat{\text{cov}}_m$ Empirical coverage in bin m

p Probability / dropout rate / bootstrap prob.

y Observed outcome or target value

QS_τ Quantile score (pinball loss)

K Number of quantile levels or ensemble heads

$\{\tau_k\}_{k=1}^K$ Set of quantile levels

Statistical Analysis

d Cohen's d effect size

δ Cliff's delta effect size

n, N Sample size, total observations

M Number of bins (calibration)

B Number of bootstrap resamples

μ_1, μ_2 Sample means for two groups

s_1^2, s_2^2 Sample variances for two groups

n_1, n_2 Sample sizes for two groups

$\mathbb{E}[\cdot]$ Expected value operator

$\mathbb{I}\{\cdot\}$ Indicator function

$\mathcal{N}(0, \sigma^2 I)$ Multivariate normal

Experimental Design

σ Environmental noise level

ε Epsilon (exploration parameter)

A.2 Abbreviations

UQ Uncertainty Quantification

RL Reinforcement Learning

DRL Deep Reinforcement Learning

DQN Deep Q-Network

QR-DQN Quantile Regression DQN

MC Dropout Monte Carlo Dropout

CRPS Continuous Ranked Probability Score

ACE Average Calibration Error

WIS Weighted Interval Score

PICP Prediction Interval Coverage Probability

CDF Cumulative Distribution Function

MLP Multi-Layer Perceptron

SB3 Stable Baselines3

NLL Negative Log-Likelihood

SD Standard Deviation

UCB Upper Confidence Bound

CI Confidence Interval

PMLR Proc. Machine Learning Research

ICML Int'l Conf. Machine Learning

NeurIPS Neural Information Processing Systems

AAAI Assoc. Advancement of AI

IEEE Institute of Electrical Engineers

A.3 Key Terminology

Aleatoric Uncertainty Irreducible uncertainty inherent in the environment or data generation process.

Epistemic Uncertainty Reducible uncertainty in model parameters due to limited training data.

Calibration Degree to which predicted confidence levels match empirical frequencies.

Coverage Empirical frequency that prediction intervals contain true values.

Ensemble Disagreement Variance across multiple models, used as an uncertainty measure.

Episode Complete sequence of state-action-reward interactions ending in terminal state.

Noise Paradox Counterintuitive finding that moderate noise can improve UQ quality.

Proper Scoring Rule Scoring function encouraging honest probabilistic forecasts.

Quantile Regression Method for estimating conditional quantiles of response distribution.

Sharpness Degree to which prediction intervals are narrow (informative).

Bootstrap Mask Binary sampling indicator determining which ensemble heads update for each experience.

Pinball Loss Loss function for quantile regression, equivalent to quantile score.

Post-hoc Calibration Adjustment of predictions after training to improve calibration.

Overestimation Bias Systematic tendency of Q-learning to overestimate action values.

B Supplementary Experiments

B.1 Task Performance (Sanity Check)

Across all experimental configurations, the training processes of the three methods showed stable convergence. Figure 3 illustrates the best-evaluation reward across different noise levels, and the aggregated results are summarized in Table 7.

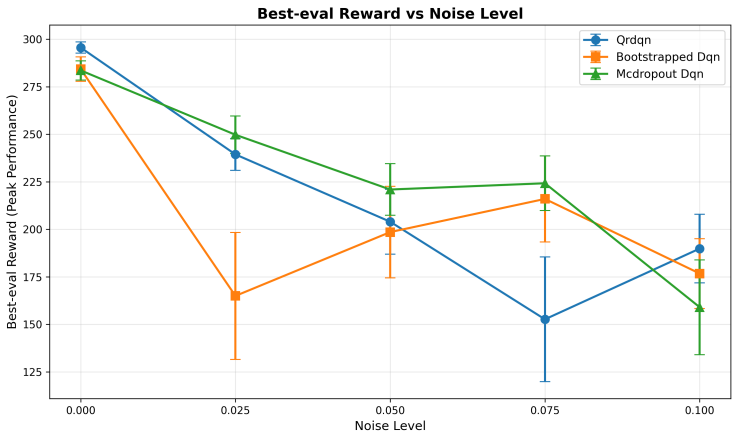

Fig. 3: Best-evaluation reward across different noise levels. The horizontal axis denotes noise σ, and the vertical axis denotes best-eval reward. Points represent means over 10 seeds, and error bars indicate the 95% confidence interval.

Table 7: Best-evaluation reward (mean ± standard deviation) for each method across five noise levels ($n = 10$ seeds per cell).

Method	$\sigma = 0.0$	$\sigma = 0.025$	$\sigma = 0.05$	$\sigma = 0.075$	$\sigma = 0.1$
QR-DQN	295.6 ± 4.8	239.4 ± 13.6	204.0 ± 27.6	152.7 ± 52.9	189.9 ± 29.1
Bootstrapped DQN	284.3 ± 10.4	165.0 ± 61.4	198.5 ± 38.9	216.0 ± 36.6	176.7 ± 29.6
MC Dropout DQN	283.6 ± 8.1	249.7 ± 15.9	220.9 ± 21.9	224.2 ± 23.2	159.0 ± 40.2

B.2 Statistical Tests for UQ Quality

To support the comparisons in Section 4.1, we conducted Mann–Whitney U tests between each pair of methods across all metrics. Effect sizes are reported as Cohen's d.

Table 8: Pairwise statistical comparisons between UQ methods on CRPS, WIS, ACE, and Coverage ($n = 50$ runs per method). Values are reported as gain % / p-value / Cohen's d. "pp" indicates percentage points for coverage.

Comparison	**CRPS**	**WIS**	**ACE**	**Coverage**
QR-DQN vs Bootstrapped	31% / $p < 0.001$ / $d = 0.92$	55% / $p < 0.001$ / $d = 1.45$	71% / $p < 0.001$ / $d = 3.21$	+52 pp / $p < 0.001$ / $d = 2.87$
QR-DQN vs MC Dropout	45% / $p < 0.001$ / $d = 1.23$	60% / $p < 0.001$ / $d = 1.78$	69% / $p < 0.001$ / $d = 2.95$	+50 pp / $p < 0.001$ / $d = 2.64$
Bootstrapped vs MC Dropout	– / $p = 0.067$ / ns	– / $p = 0.032$ / $d = 0.19$	– / $p = 0.089$ / ns	– / $p = 0.245$ / ns

All tests used a significance threshold of $p < 0.05$. Results indicate medium to large effects for QR-DQN against both alternatives, while Bootstrapped DQN and MC Dropout DQN did not significantly differ on most metrics.

B.3 QR-Bootstrapped DQN (Exploratory Attempt)

We experimented with a QR-Bootstrapped DQN variant, aiming to combine the strengths of distributional estimation and bootstrapped ensembles. However, the method was unstable and did not outperform either baseline. For transparency, aggregated results across noise levels are reported in Table 9.

Table 9: Performance of QR-Bootstrapped DQN across noise levels (mean ± standard deviation, $n = 10$ seeds per cell). Results indicate lower reward and inconsistent uncertainty quality compared to baselines.

Noise σ	Reward	CRPS	WIS	ACE	Coverage (90%)
0.00	227.6 ± 46.8	15.1 ± 7.9	178.7 ± 103.6	0.20 ± 0.20	70 ± 20
0.05	200.8 ± 58.7	14.9 ± 4.6	179.0 ± 74.6	0.10 ± 0.20	70 ± 20
0.10	129.3 ± 61.2	20.5 ± 4.0	256.1 ± 78.4	0.30 ± 0.20	60 ± 20

B.4 Noise Effects (Coverage Only)

CRPS, WIS, and ACE per-noise results are already reported in the main text (Tables 3–5). Here we provide complementary **Coverage** values for the 90% prediction interval.

Table 10: Coverage at the 90% prediction interval across noise levels (in %).

Method	$\sigma = 0.0$	$\sigma = 0.025$	$\sigma = 0.05$	$\sigma = 0.075$	$\sigma = 0.1$
QR-DQN	92 ± 8	86 ± 13	87 ± 14	80 ± 16	73 ± 16
Bootstrapped DQN	8 ± 7	33 ± 20	38 ± 18	36 ± 26	41 ± 18
MC Dropout DQN	43 ± 12	24 ± 11	23 ± 10	28 ± 15	29 ± 14

B.5 Noise Paradox (Bootstrapped DQN)

To further examine the irregular trend of Bootstrapped DQN under noise, we evaluated additional noise levels in the range $\sigma = 0.020$–0.030. Table 11 reports the calibrated CRPS across these settings ($n = 10$ seeds per cell).

Table 11: Calibrated CRPS for Bootstrapped DQN at additional fine-grained noise levels.

Noise σ	0.020	0.022	0.025	0.027	0.030
CRPS	47.0 ± 48.7	11.8 ± 4.1	13.9 ± 6.4	23.5 ± 3.5	21.2 ± 16.6

B.6 Calibration Effects (Numerical Details)

Absolute differences (post–pre). Values are mean ± SD aggregated across five noise levels ($n = 50$ per method). Negative values for CRPS/ACE indicate improvement; coverage is in percentage points.

Table 12: Aggregated calibration effects (absolute differences).

Method	ΔCRPS	ΔACE	ΔCoverage (pp)
QR-DQN	-1.1 ± 7.5	-0.14 ± 0.17	+20 ± 23
Bootstrapped DQN	-5.5 ± 9.8	-0.21 ± 0.17	+28 ± 21
MC Dropout DQN	-5.7 ± 8.0	-0.22 ± 0.12	+30 ± 14

Per-noise percentage gains. Values are means over 10 seeds per cell.

Table 13: Per-noise calibration gains (percentage).

Method	$\sigma = 0.0$	$\sigma = 0.025$	$\sigma = 0.05$	$\sigma = 0.075$	$\sigma = 0.1$
CRPS reduction (%)					
QR-DQN	-28.4	4.5	12.6	2.8	-6.0
Bootstrapped DQN	14.5	8.3	12.7	10.0	10.5
MC Dropout DQN	9.6	17.2	14.9	12.9	9.7
ACE reduction (%)					
QR-DQN	36.7	28.6	52.7	45.9	-55.8
Bootstrapped DQN	6.4	32.0	33.0	37.7	39.6
MC Dropout DQN	39.8	31.3	33.1	27.8	23.7
Coverage change (pp)					
QR-DQN	30	30	20	16	3
Bootstrapped DQN	10	30	34	33	36
MC Dropout DQN	35	31	32	27	23